

Binary Cross Entropy Loss $L = -y \ln(p) - (1-y) \ln(1-p)$

Derivative:

$$\frac{dL}{dp} = \frac{d}{dp} (-y \ln(p) - (1-y) \ln(1-p))$$

$y = \text{constant}$

$$= -y \frac{d}{dp} \ln(p) - (1-y) \frac{d}{dp} \ln(1-p)$$

$$p = \sigma(x)$$

$\sigma(x) = \text{sigmoid function}$

$$= -y \cdot \frac{1}{p} - (1-y) \cdot \frac{1}{1-p} \cdot \frac{d}{dp} (1-p)$$

$$= -\frac{y}{p} - \frac{1-y}{1-p} (0-1)$$

$$= -\frac{y}{p} + \frac{1-y}{1-p}$$

$$= \frac{-y(1-p) + (1-y)p}{p(1-p)}$$

$$= \frac{-y + py + p - py}{p(1-p)}$$

$$\frac{dL}{dp} = \frac{p-y}{p(1-p)}$$

Sigmoid output -

$$p = \sigma(x) = \frac{1}{1+e^{-x}}$$

$$p = (1+e^{-x})^{-1}$$

$$\frac{dp}{dx} = \frac{d}{dx} (1+e^{-x})^{-1}$$

$$= -1 \cdot (1+e^{-x})^{-2} \cdot \frac{d}{dx} (1+e^{-x})$$

$$= \frac{e^x}{(1+e^x)^2}$$

$$= \frac{e^x}{(1+e^x)^2} \cdot \frac{1}{(1+e^x)}$$

$$= \frac{1+e^x - 1}{1+e^x} \cdot p$$

$$= \left(\frac{1+e^x}{1+e^x} - \frac{1}{1+e^x} \right) p$$

$$= (1-p) p$$

$$\frac{dP}{dz} = (1-p) p$$

Apply Chain Rule $\rightarrow$

$$\frac{dL}{dz} = \frac{dL}{dP} \times \frac{dP}{dz}$$

$$= \frac{P-y}{P(1-P)} \times (1-p) \times p$$

$$= P-y$$

Ans: $\frac{dL}{dz} = P-y$